🐛 Bug Report: Configuration Migration Failure During Upgrade (v3.24 → v3.28)

Summary

After upgrading from v3.24 to v3.28, the OpenClaw gateway failed to start due to corrupted configuration file generated by the migration script. The `openclaw doctor --fix` command was also unable to repair the corrupted JSON, requiring manual intervention.

Environment

- **OpenClaw version**: v2026.3.28 (upgraded from v2026.3.24)
- **Node version**: v24.14.0 (via nvm)
- **OS**: Ubuntu 24.04 (Linux 6.17.0-19-generic x64)
- **Install method**: npm global install (`npm i -g openclaw`)

Steps to Reproduce

1. Have OpenClaw v3.24 installed with feishu channel configured
2. Upgrade to v3.28: `npm i -g openclaw@latest`
3. Run `openclaw gateway start` or `openclaw doctor`

Actual Behavior

Error 1: Invalid JSON Generated by Migration

The migration script produced invalid JSON in `~/.openclaw/openclaw.json` :

```
 "channels": {
"feishu": {
    "enabled": true,
    ...
  }      "enabled": true    // ← Invalid JSON: missing comma, duplicate
key
}
```

This caused JSON parsing failures and gateway startup failure.

Error 2: Config Validation Failure

After manually fixing the JSON structure, the config validation failed with:

```
 error: config validation failed: channels.feishu.accounts.main: invalid
config: must not have additional properties
```

Root Cause: Breaking Config Schema Change

Old format (v3.24):

```
 "channels": {
  "feishu": {
    "enabled": true,
    "accounts": {
      "main": {
        "appId": "cli_xxx",
        "appSecret": "xxx",
        "botName": "My Bot"  // ← Deprecated field
      },
      "default": {
        "dmPolicy": "pairing"  // ← Deprecated account
      }
    }
  }
}
```

New format (v3.28):

```
"channels": {
 "feishu": {
   "enabled": true,
   "dmPolicy": "pairing",           // ← Moved to top level
   "groupPolicy": "allowlist",      // ← Moved to top level
   "accounts": {
     "main": {
       "appId": "cli_xxx",
       "appSecret": "xxx",
       "name": "My Bot"             // ← Renamed from botName
       // Only allowed: appId, appSecret, name, domain, enabled
     }
   }
 }
}
```

Key changes:

1. `botName` renamed to `name`
2. `default` account key removed
3. Policy fields (`dmPolicy` , `groupPolicy`) moved from `accounts.default` to top level
4. Strict schema: only `appId` , `appSecret` , `name` , `domain` , `enabled` allowed in accounts

Expected Behavior

1. **Migration should produce valid JSON** - The migration script should never output syntactically invalid JSON
2. **Automatic rollback on failure** - If migration fails, restore from backup automatically
3. **Clear error messages** - `doctor --fix` should detect and repair schema migration issues
4. **Documentation of breaking changes** - Release notes should document config schema changes

Workaround (Manual Fix)

1. Locate backup file: `~/.openclaw/openclaw.json.bak.2` (or earlier backup)
2. Manually update config to new schema format
3. Run `openclaw doctor --fix` to apply remaining changes

Evidence

Backup Files Created

```
 -rw------- 1 xmliao xmliao 4058 3月 30 14:12 openclaw.json.bak
-rw------- 1 xmliao xmliao 3790 3月 30 14:12 openclaw.json.bak.1
-rw------- 1 xmliao xmliao 3909 3月 24 07:23 openclaw.json.bak.2  #
Working config from v3.24
-rw------- 1 xmliao xmliao 4058 3月 30 14:08 openclaw.json.clobbered.
2026-03-30T06-08-06-074Z
```

Clobbered File Content

The `.clobbered` file shows invalid JSON structure where the migration script incorrectly merged/transformed the feishu channel config.

Impact

Aspect	Description
Severity	High - Gateway fails to start after upgrade
User experience	Requires manual config editing and understanding of schema changes
Affects	Users upgrading from v3.24 (or earlier) with feishu channel configured

Suggested Fixes

Short-term

1. Improve migration script to handle all edge cases
2. Add JSON validation after migration
3. If validation fails, restore from `.bak` and log clear error message

Long-term

1. Add config version field for explicit versioning
2. Implement reversible migrations
3. Add `openclaw config migrate --from <version>` command
4. Include config backup/restore in `doctor --fix`

Related Files

- `~/.openclaw/openclaw.json` - Main config file
- `~/.openclaw/openclaw.json.bak.*` - Automatic backups
- `~/.openclaw/openclaw.json.clobbered.*` - Corrupted migration output

Reporter: geogleo
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Upgrade path: v2026.3.24 → v2026.3.28